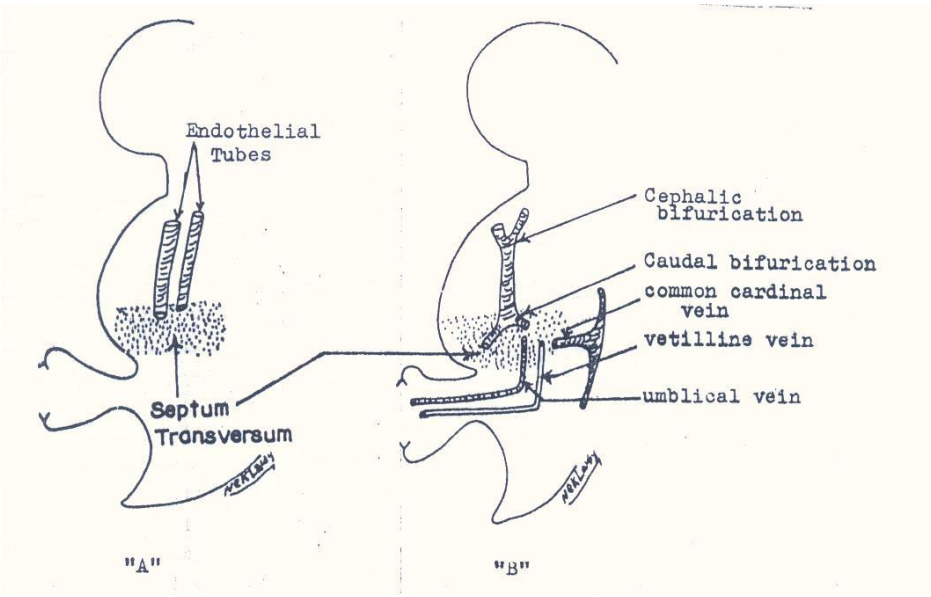
Development of the heart

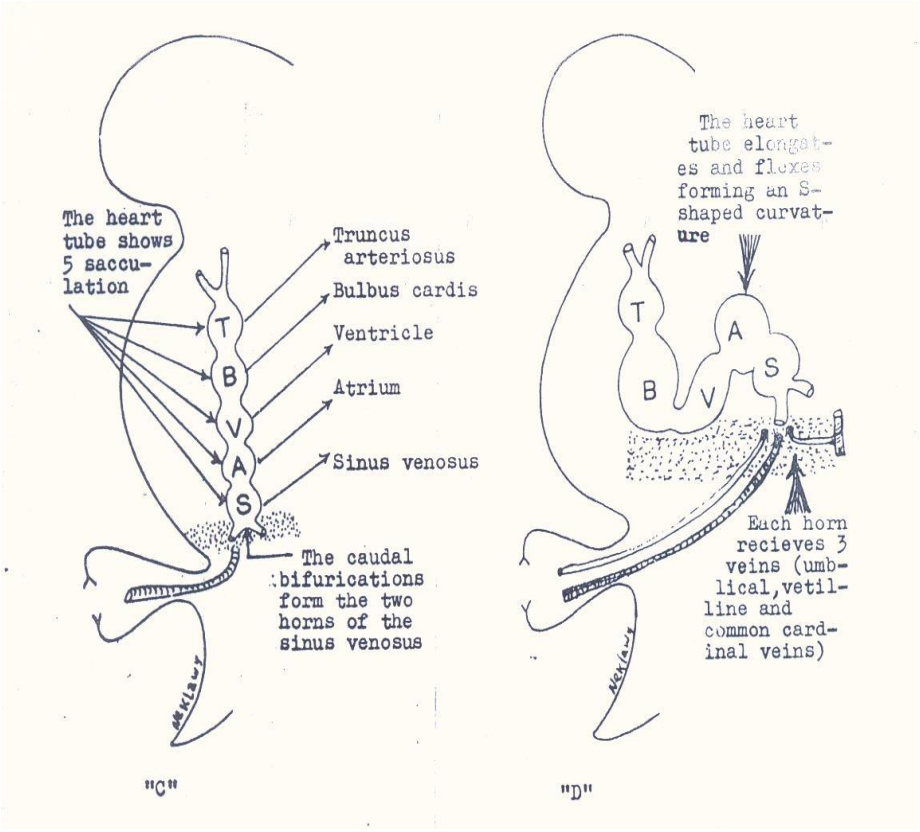
The angioblastic cells in the cardiogenic swelling are arranged into two endothelial tubes which unite at their middle segments leaving a cephalic and caudal bifurcations.

The cephalic bifurication coummunicates with cephalic ends of the dorsal aortae, The caudal bifurcation sinks into the septum transversum to communicate with the umbilical, vetilline and common cardinal veins on each side.

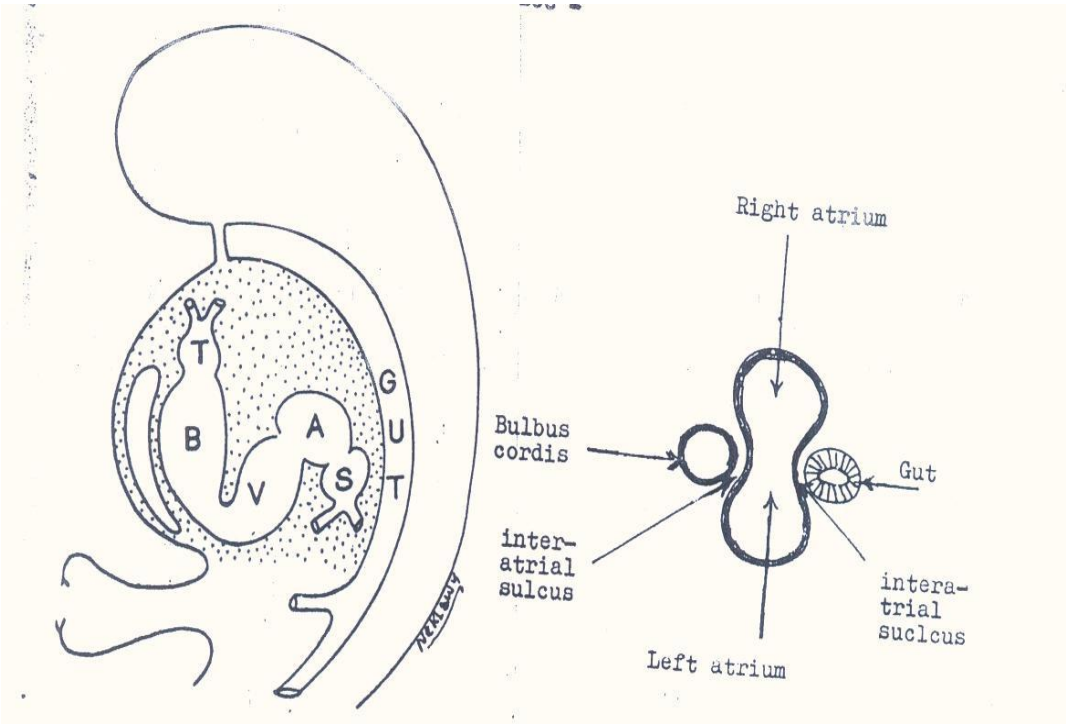
The heart tube begins to show five sacculations. The most cephalic one is the truncus arteriosuswhich is followed by the bulbus cordis, ventricle, atrium and then the sinus venosus. The latter takes the caudal bifurcation as two horns..Each horn of the sinus venosus receives thee veins, umbilical, thevetilline and common cardinal veins.

By further development, the sacculated heart tube elongates as the cardiogenic swelling can't accommodate this elongation.The tube will flex dorsally resulting in an S-shaped curvature.Thus , the position of he saccules will be changed where the ventricle and the atrium became behind the bulbus cordis while that of the sinus venosus becomes dorsal to the atrium





Formation of the heart tube



Development of the atrium:

The developing atrium lies between the bulbus cordis ventrally and the gut dorsally. Then the atrium begins to enlarge in all directions but this enlargement is arrested by the bulbus cordis ventrally and the gut dorsally. Thus, a sulcus appears on the external surface of the atrium dividing it externally into two atria (Right and left atrium). This is the interatrial sulcus. Up to this stage, the two atria are freely communincated internally (both forming a common chamber). The common atrial chamber is still in communication dorsally with the sinus venosus. The latter is then shifted out of the mid-line so that it opens into the back of the right atrium by an ellipitical orifice (sinoatrial orifice) which is gurded by a valve. This sinal valve has 2 vertical cusps, (right and left).

The sinus venosus has right and left horns which communicate with each other by a transverse coummicating duct.As the blood is shifted to to the right, it results into great enlargement of the right horn and diminuation of the left horn. Later, the sinus venosus and its right horn will be absorbed into the back of the right atrium.

The right atrium is therefore derived partly from sinus venosus and partly from the primitive atrium. By absorption of the sinus venosus, its opening will be closed with disappearance of the left cusp while the right cusp remains as what is called (crista terminalis) whichextends as a vertical ridge into the atrial cavity (from the opening of the future superior vena cava to the inferior vena cava). This crista terminalis is the mark which diffeniate the sinal part from that of the atrial part. The sinal part is smooth and lies behind the crista terminalis while the atrial part is rough and found infront of the crista terminalis. Roughness is due to the development of the muscular ridges or trabeculae into the atrial cavity known as musculi pectinati.

As the left horn becomes rudiment, it remains as the oblique vein of marshal which runs on the back of the left atrium. This vein opens into the coronary sinus which originates from the transverse coummunicating duct between the 2 horns. The coronary sinus extends along the back of the atrio- ventricular groove to open into the right atrium just on the left of the inferior vena caval opening. Each of the two openings is guarded by a valve which originate from the lower end of the crista terminalis.

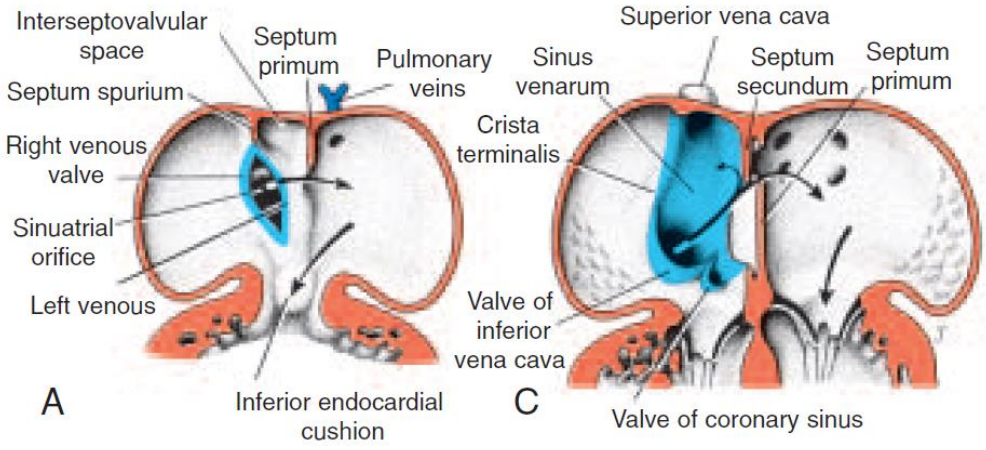


Figure 11.12 Ventral view of coronal sections through the heart at the level of the atrioventricular canal to show development of the venous valves. **A.** 5 weeks. **B.** Scanning electron micrograph of a similar-staged mouse heart showing initial formation of the septum primum; septum spurium is not visible. Note the atrioventricular canal (*arrow*). **C.** Fetal stage. The sinus venarum (*blue*) is smooth walled; it derives from the right sinus horn. *Arrows*, blood flow. **D.** High magnification of the interatrial septum (*arrows*) of a mouse embryo at a stage similar to **C.** The foramen ovale is not visible.

Formation of the inter-atrial septum:

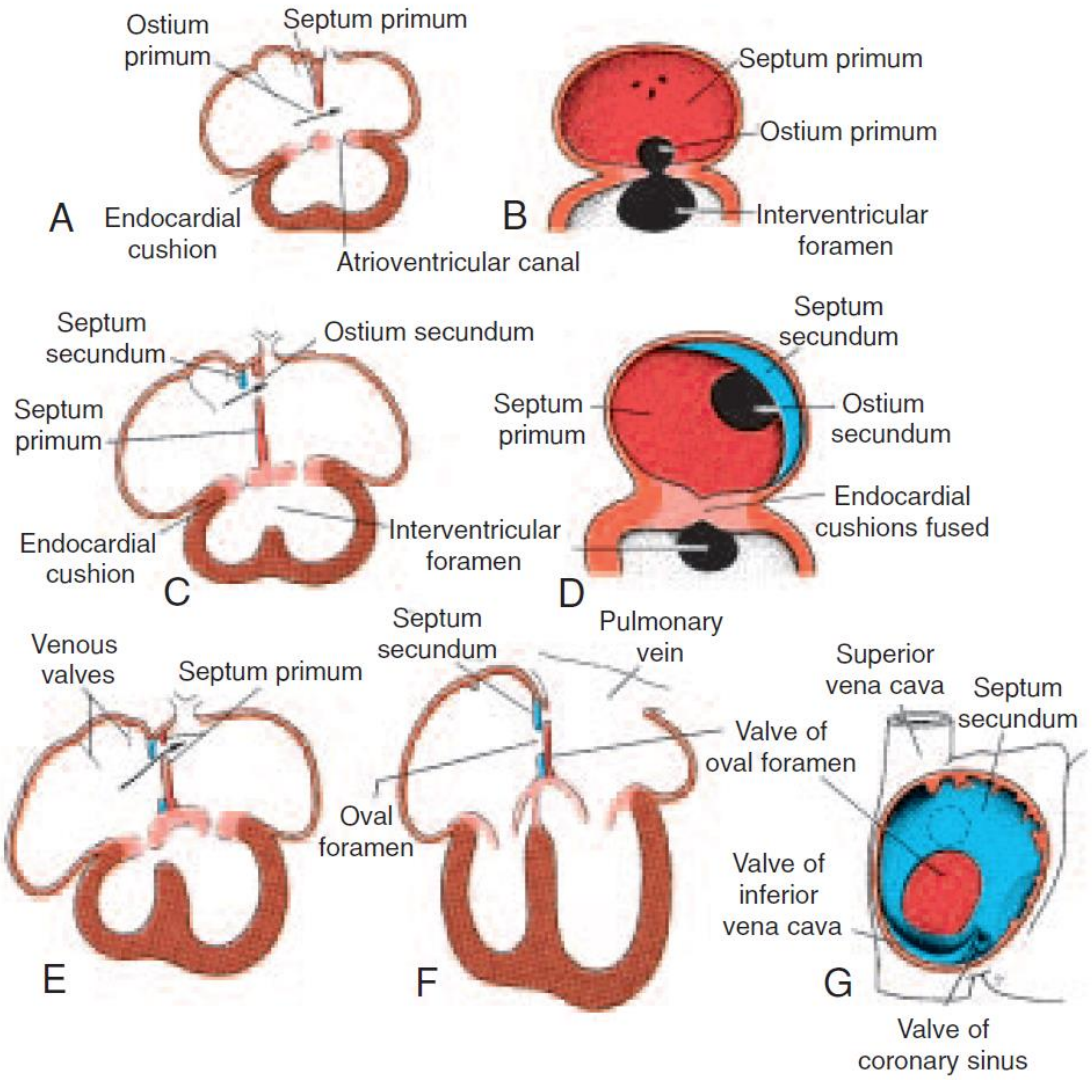
The atrium is originally a single chamber which , then, partially subdivided into right and left atrium by an external sulcus (inter-atrial sulcus) as a result of a constrication produced by the bulbus cordis infront and gut behind. However, the 2 atria are freely communicating with each other.

Later on, a septum will appear (opposite the interatrial sulcus) and known as the primary septum or (septum primum). It grows downwards into the common chamber to divide it partially into right and left atrium. The lower margin of the sptum primum is free and being crescentic in its shape. Thus it leaves an orifice termed as the foramen primum Which

allows free communication between the 2 atria. Thus the foramen primum is bounded above by the lower crescentric margin of the septum primum and below by a growing sub-endocardial cushion atrio-ventricular cushion which extends anter0- posteriorly at the atrioventricular canal. The atrioventricular cushion appears first as dorsal and ventral mass of embryonic connective tissue which later, brought into contact with each other by their own growth to form a common mass dividing the atrio- ventricular canal.

Then, he septum primum proceeds downwards until it fuses with atrio-ventricular cushion closing the foramen primum and thus the 2 atria are completely separated. However , a degenerative process takes place in the upper part of the septum primum leaving this septum with an upper crescentic free margin. The new orfice is bounded below by the upper free margin of the septum primum. This orifice is known as the foramen secondum.Then, a new septum is growing to the right of the septum primum. This is the septum secondum which has a lower free crescentric margin.

As the septum secondum grows downwards on the right of the remaining part of the septum primum, it does not form a complete partition. Thus, its lower concaved margin leaves an oval aperature which is the foramen oval (the foramen oval is an aperature into the septum secondum. It is covered on the left by the remaining part of the septum primum. The latter acts as a loose valvular flap which allows passage of blood from the right to the left but prevents the flow of blood in the reverse direction. This flap is commonly known as the (valvula foramins ovalis) rather than the name of septum primum. It constituates what is in the adult anatomy is called (fossa ovalis), while the the concaved margin of the septum secondum constitutes the (annulus ovalis).



Stages of development of the inter-atrial septum

Formation of the interventricular septum

From the apex of the common ventricular chamber, a muscular septum grows upwards forming the muscular part part of the inter-ventricular septum. I It divides the common chamber into two chambers, however they are freely communicate with each other as this muscular septum is still incomplete. It leaves an aperature which is the interventricular foramen. This foramen wil be closed by a membrane originating from the connective tissue of the atrioventricular cushion as will as the muscular interventriculat septum. Thus , the interventricular septum is partly muscular and partiy membranous.

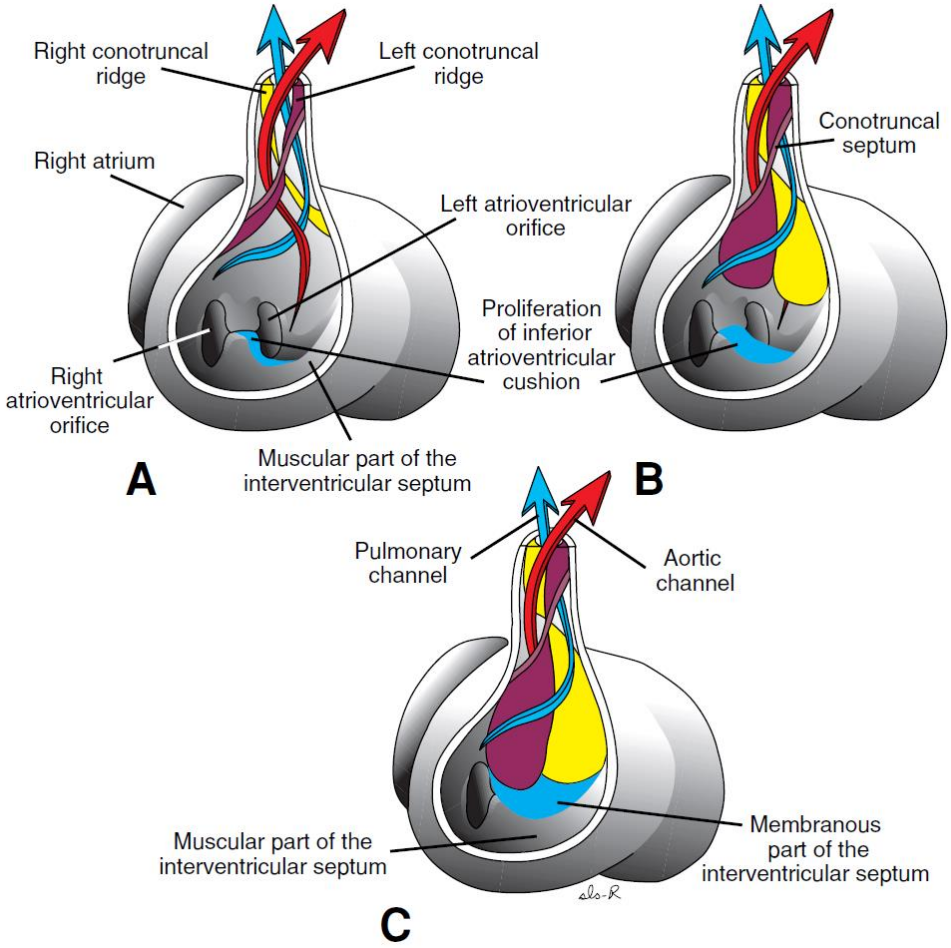
Fate of the bulbus cordis and truncus arteriosus

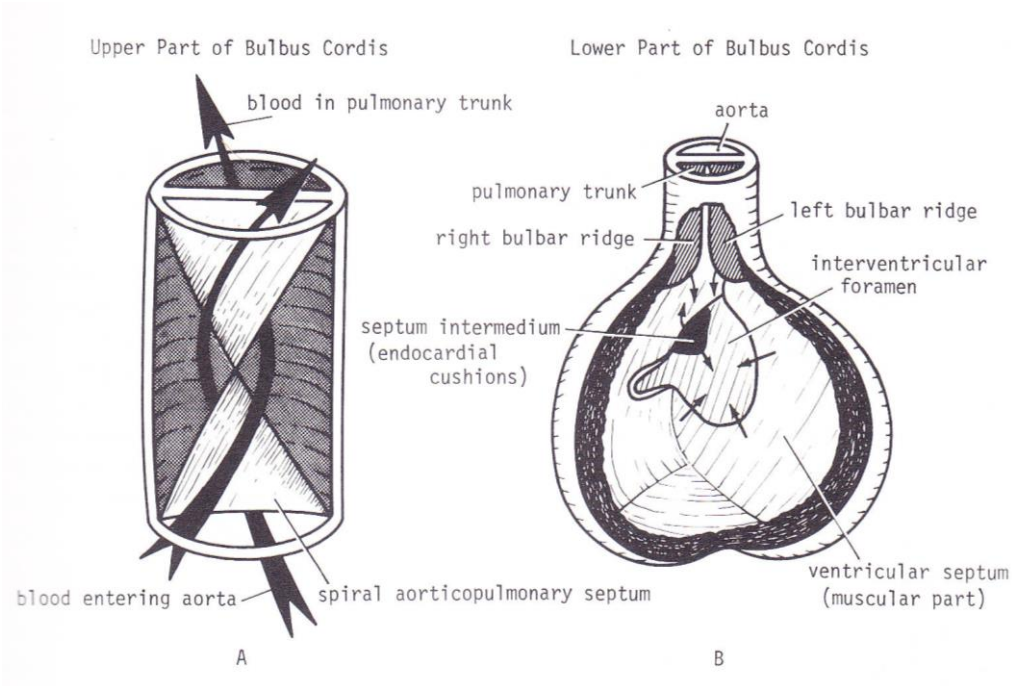
The proximal part of the bulbus cordis will be absorbedpartly into the left ventricle forming the vestibule of the ascending aorta and partly into the right ventricle to form the infundibulum of the pulomonary trunk. The distal part of the bulbus cordis and the whole of the truncus arteriosus show 2 vertical sulci (right and left) on their external surface, Thus the common tube is divided externally into 2 tubes one dorsal (future ascending aorta) and one is ventral (future pulmonary trunk). Internally the 2 tubes are still freely communicating with each other. Later, a septum willappear internally(corresponding in position to the external sulci). To divide the interior of the tube into 2 tubes. The proximal end of this septum (at the bulbus cordis) meets 2 great masses (right and left cushions) which appear into the cavity of the distal end of the bulbus cordis and each extends medially to fuse with each other.

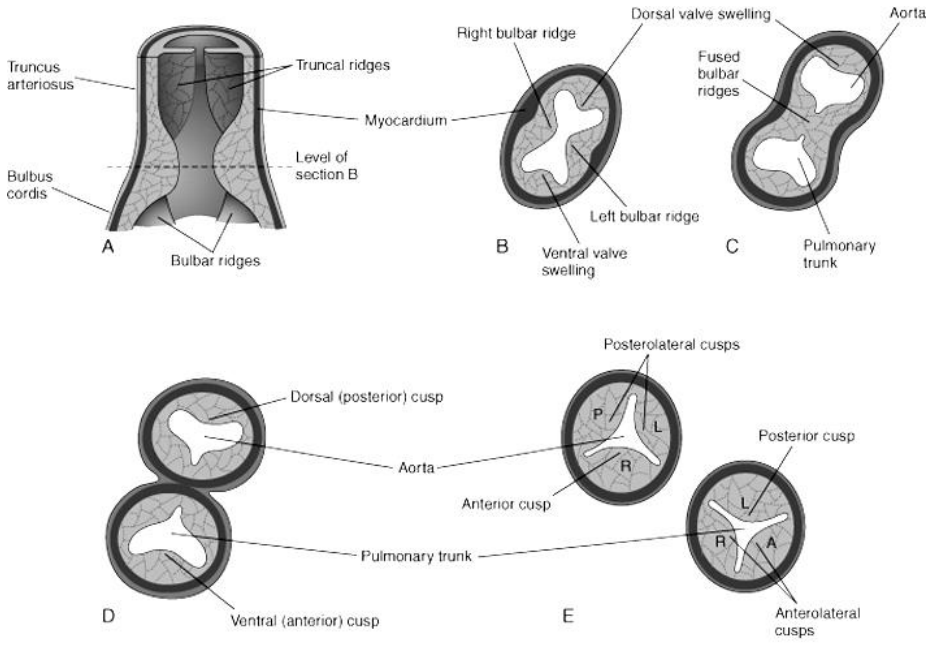
Then, 2other small cushions (ventral and dorsal) grow into this cavity but never fuse with each other. Meeting of the septum with the great cushions will divide it into 2 haves (dorsal and ventral). Thus , the ventral tube has a small cushion ventrally and half of the great cushions dorsally, while the dorsal tube has a dorsal cushion dorsally and half of great cushions ventrally.

As the dorsal tube will be the ascending aorta and the ventral one will be the pulmonary trunk. Thus, up to this stage the aorta has 2 ventral cushions and one dorsal. While the pulmonary trunk has one ventral and 2 dorsal cushions. By further

development, the ascending aorta and the pulmonary trunk rotat 45 degree. So that a positional changes will be resulated. Thus, the aorta has one ventral and 2 dorsal , while the pulmonary has 2 ventral and one dorsal cushions. These cushions will be excavated from the upper surface to o form the aortic and pulmonary cusps (3 for each). Furtherly, the aortic and pulmonary tubes are twisted in a spiral fashion, so that the sulcus is locked dorsally in the middle of the distance, and then to the right while the right sulcus is locked ventrally and then to the left in the upper part of the distance.







Formation of the semilunar valves

DEVELOPMENT OF CONDUCTING SYSTEM

Initially the **pacemaker** of the heart is located in the caudal part of the left heart tube. After appearance of the sacculations in the heart tube, it is believed that parts of all interchamber rings (sinoatrial, atrioventricular, ventriculobulbar and truncobulbar) are involved in development of the conducting system.

A-V node, A-V bundle and bundle branches

The atrioventricular node (**A-V node**) is the first to appear as an outgrowth from the ***cells*** of the ***dorsal atrioventricular canal.*** Another source for **A-V node** and **A-V bundle** is cells from the ***left wall of sinus venosus*** found at the base of the interatrial septum. **A-V bundle** is a subsequent extension from the **A-V node** which passes distally behind the dorsal endocardial cushion, then along the dorsal part of the muscular interventricular septum where it gives rise first to **left** then to **right bundle branches**. Bundle branches are distributed throughout the ventricular myocardium.

S-A node

The sinoatrial node (**SAN)** is the last of the conducting system to appear (week 5) from the ***right wall of the sinus venosus*** **SAN** and will be located near the entrance of the superior vena cava.

The hear starts to beat during the 3rd week prior to development of the conducting system, and a circulation is established before a competent valvular mechanism. The conducting system is well developed before nerves enter the heart. At first contractions of the heart are purely muscular, but several weeks later the rhythm is modified by development and entry of the autonomic nerves.

ANOMALIES OF THE HEART

-Congenital heart defects (CHDs) are common.

- -Some CHDs are associated with early cyanosis, e.g. transposition of great arteries, persistent truncus arteriosus and tetralogy of Fallot. These anomalies are characterized by right to left shunts. Other anomalies that may cause left to right shunts e.g. atrial septal defect, ventricular septal defect and patent ductus arteriosus cause late cyanosis because of the overload of the right ventricle.

1-Acardia

Agenesis of the heart is compatible with life in conjoined twins.

2-Dextrcardia

The heart and its vessels are reversed left to right as in a mirror image.

3-Atrial septal defects (ASDs)

ASDs are common anomalies more frequently in females than males. Depending on the size of the defect, manifestations vary from clinically not significant to considerable left to right shunt with overload on the right ventricle. The commonest cause of ASD is failure of closure of foramen oval.

4-Ventricular septal defects (VSDs)

- Ventricular septal defects are the most common types of CHDs (25% of CHD). VSDs are more frequent in males.
- They may occur in any part of the IV septum.
- Many small VSDs close spontaneously (30-50%) during the first year.
- Large VSD results in pulmonary hypertension and heart failure.

5-Fallot tetralogy

- It is a common abnormality.
- It is due to unequal division of TA and anterior displacement of the conotruncal septum.

-This manifests in four defects:
a)Pulmonary stenosis.
b)VSD.
c)Right ventricular hypertrophy.
d)Overriding of the aorta above VSD.

Cyanosis is an obvious sign of Fallot tetralogy.

6-Eisenmenger syndrome: is congenital heart defect shows a similar findings to those of Fallot's tetralogy but without plumomary stenosis.

7-Valve anomalies

a) Pulmonary stenosis

- Pulmonary stenosis is narrowing of the pulmonary valve.
- The right ventricle contracts at a higher pressure and less blood is pumped to the lungs to be oxygenated.

b) Aortic stenosis

- Consequently, the left ventricle pumps with higher pressure and blood flow to the body is inadequate.

8-Transposition of great arteries (TGA)

- TGA is the most common cause of cyanotic heart disease in newborn infants.
- It is often associated with other cardiac anomalies (ASD and VSD).
- Aorta arises from the right ventricle and the pulmonary trunk arises from the left ventricle.
- ASD and VSD permit change between pulmonary and systemic circulations. Without surgical correction, infants usually die within few months.
-TGA is due to defective migration of neural crest cells that are involved in formation of truncal cushions.

9-Persistent truncus arteriosus (TA)

-TA is due to failure of normal development of the aorticopulmonary septum.
- TA overrides an accompanied VSD. Single arterial trunk arises from the ventricles and supplies systemic and pulmonary circulations.
- Early cyanosis is present.

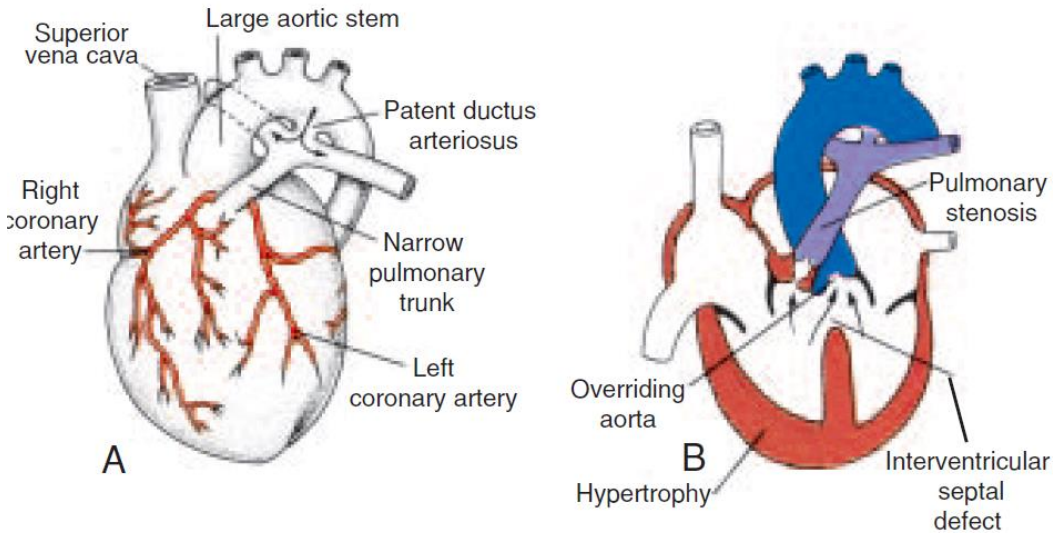


Figure 11.29 Tetralogy of Fallot. **A.** Surface view. **B.** The four components of the defect: pulmonary stenosis, overriding aorta, interventricular septal defect, and hypertrophy of the right ventricle. **C.** Ultrasound scan showing a normal heart with atria (*asterisks*), ventricles (V), and interventricular septum (*arrow*). **D.** Scan of a heart showing the characteristic features of the tetralogy, including hypertrophy of the right ventricle (*R*) and overriding aorta (*arrow*). *A*, atrium; *L*, left ventricle.

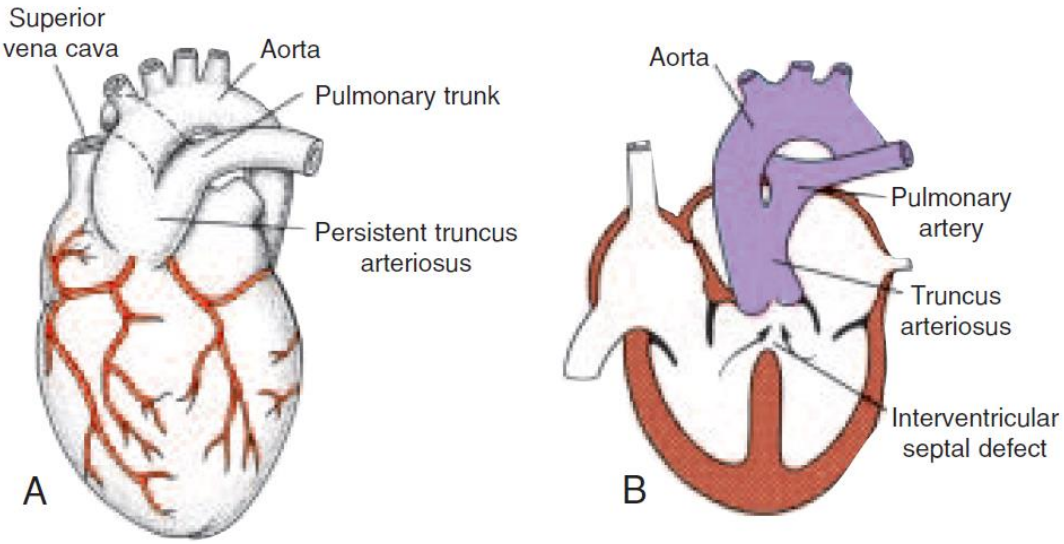
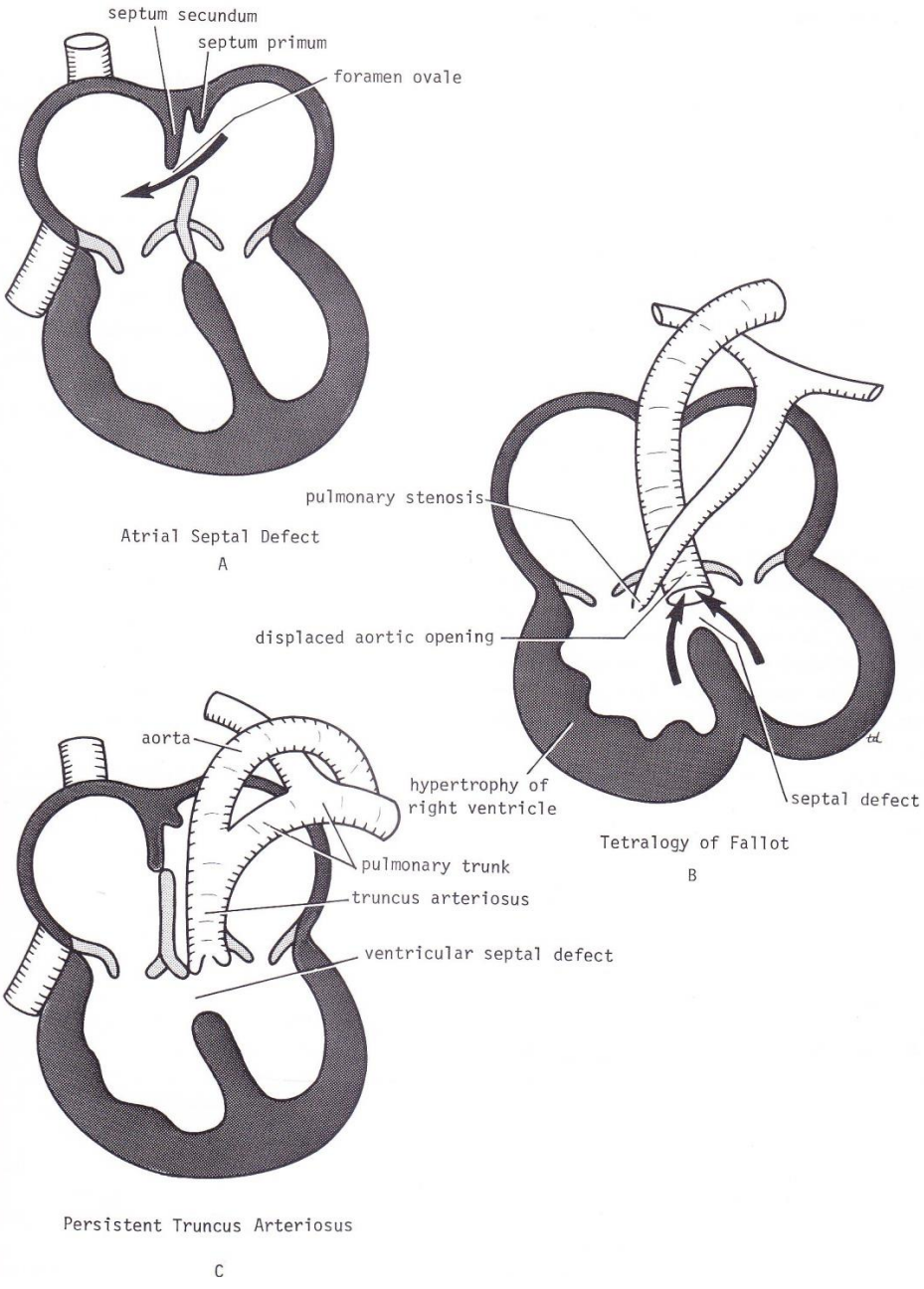


Figure 11.30 Persistent truncus arteriosus. The pulmonary artery originates from a common truncus (**A**). The septum in the truncus and conus has failed to form (**B**). This abnormality is always accompanied by an interventricular septal defect.


septum secundum
septum primum
foramen ovale
Atrial Septal Defect
A
pulmonary stenosis
displaced aortic opening
hypertrophy of
right ventricle
septal defect
Tetralogy of Fallot
B
aorta
pulmonary trunk
truncus arteriosus
ventricular septal defect
Persistent Truncus Arteriosus
C